



NurvexThink

Software, built and published.

Main Website — Design Specification

Public hub: product showcase, blog, custom-order intake, admin dashboard

Date: 2026-06-30
Status: Approved (design phase)
Owner: Fatima Abdul Raheem (CEO)

Contents

1 Overview	2
1.1 Goals	2
1.2 Non-goals (deferred)	2
2 Brand	2
3 Architecture	2
3.1 Stack	3
4 Components / Pages	3
5 Data model (Supabase)	3
5.1 Tables	3
5.2 Storage buckets	3
5.3 Row Level Security (enabled on EVERY table, default-deny)	3
6 Security	4
7 Deployment & GitHub	4
7.1 Vercel/GitHub deployment notes (researched)	4
8 Org management	4
8.1 GitHub org (nurvexthink)	4
8.2 Supabase org (nurvexthink)	4
9 Performance & quality (researched best practices)	5
9.1 Next.js	5
9.2 React Three Fiber / Three.js	5
9.3 Supabase / Postgres	5
10 Testing	5
11 Build order (high level)	5
12 Open items / future phases	5

1 Overview

NurvexThink is a software company that builds and publishes software products, takes custom software orders on demand, and writes blog posts about its work. This spec covers the **main website** — the public hub that showcases products (with links to their live apps), hosts the blog, captures custom-software leads, and provides an admin dashboard for the team to manage content.

Out of scope (separate future specs): the individual software products themselves. Each product is its own repository and deployment, linked to from this site.

1.1 Goals

- Premium, dark/metallic brand presence with signature 3D visuals.
- Showcase published software with links to live apps.
- Blog (created/edited via an admin dashboard — no coding required to publish).
- Capture custom-software leads via a simple form.
- \$0 to launch (free tiers), with a clear paid-upgrade path.

1.2 Non-goals (deferred)

- Client accounts / order-tracking dashboard (lead form only at launch).
- Online payments (Stripe) — later phase.
- Multi-language / i18n.

2 Brand

Standardized name: **NurvexThink** (display) / `nurvexthink` (lowercase: domains, repos, handles). Theme derived from the logo (navy/slate “N” + metallic silver “T” on black).

Token	Value	Use
--bg	#0A0A0B	near-black background
--primary-1	#1E2A44	navy/slate (gradient start)
--primary-2	#2D3A5C	slate (gradient end)
--accent-1	#C0C5CE	metallic silver (gradient start)
--accent-2	#E8EBF0	bright silver (gradient end)

Fonts: Space Grotesk / Sora (headings), Inter (body).

Vibe: dark-mode-first, glassmorphism panels, metallic gradients on hover, 3D “NT” hero.

3 Architecture

Approach A (chosen): Next.js + Supabase on Vercel. No separate backend server.

```
[ Browser ]
  | HTTPS
[ Next.js 15 App Router on Vercel ]
  | - React Server Components (data fetching, SEO)
  | - Client Components (3D, interactivity) at the leaves
  | - Route Handlers / Server Actions (form mutations)
  |
[ Supabase Cloud ]
  - Postgres (data) - Auth (admin login)
  - Storage (images) - Edge Functions (custom logic, if needed)
  - Row Level Security (RLS) on every table
```

3.1 Stack

Layer	Choice
Language	TypeScript
Framework	Next.js 15 (App Router) + React
Styling	Tailwind CSS + shadcn/ui
3D	React Three Fiber + drei + Three.js (Spline optional for no-code scenes)
Animation	Framer Motion (+ GSAP for advanced sequences)
Backend	Supabase (Postgres, Auth, Storage, Edge Functions)
Hosting	Vercel (site) + Supabase Cloud (backend)
Deploy	Public GitHub repo + GitHub Actions + Vercel token

4 Components / Pages

Route	Type	Purpose
/	public	Home: 3D metallic “NT” hero, featured products, CTA
/products	public	Grid of published software
/products/[slug]	public	Product detail + link to live app + repo
/blog	public	Blog list
/blog/[slug]	public	Blog post
/order	public	“Request custom software” lead form
/about	public	Company + team
/contact	public	Contact info
/admin	protected	Dashboard: manage products, blog posts, view leads
/admin/login	public	Admin auth (Supabase)

5 Data model (Supabase)

5.1 Tables

- `profiles` — `id` (= `auth.users.id`), `email`, `full_name`, `role` (`owner` / `admin`), `created_at`
- `products` — `id`, `slug` (`unique`), `name`, `summary`, `description`, `cover_image`, `live_url`, `repo_url`, `tags`, `featured`, `published`, `timestamps`
- `blog_posts` — `id`, `slug` (`unique`), `title`, `excerpt`, `content`, `cover_image`, `author_id`, `status` (`draft` / `published`), `published_at`, `timestamps`
- `orders` — `id`, `name`, `email`, `company`, `project_type`, `budget`, `details`, `status` (`new` / `contacted` / `closed`), `created_at`

5.2 Storage buckets

- `product-images` (public read, admin write)
- `blog-images` (public read, admin write)

5.3 Row Level Security (enabled on EVERY table, default-deny)

- `products/blog_posts`: public SELECT where published; admins full access.
- `orders`: public INSERT only (with validation); admins SELECT/UPDATE; no public read.
- `profiles`: a user reads/updates own row; admins read all.
- Index every column referenced in an RLS policy (e.g. `published`, `status`, `author_id`).

6 Security

- RLS on every public-schema table, no exceptions (Supabase is default-deny).
- `service_role` key is server-side only — never in client code or the repo; stored as a Vercel / GitHub Actions secret.
- `anon` key is public-safe because RLS protects the data.
- Lead form: validate input server-side; add basic anti-spam (honeypot / rate limit).
- Admin routes guarded by Supabase Auth + middleware; `/admin` checks `profiles.role`.

7 Deployment & GitHub

- **Repo:** `nurvexthink/nurvexthink-web` (public).
- **Vercel project:** named independently (e.g. `nurvexthink`); custom domain later.
- **Deploy:** GitHub Actions builds and deploys with a Vercel token on merge to `main`. This sidesteps Vercel Hobby's org-private-repo + commit-author restrictions, so all three admins' merged work deploys for free.
- **Env vars / secrets** live in Vercel + GitHub Actions, never in the repo.
- **Branch protection:** `main` requires a PR + 1 review; no direct pushes.
- **Commit identity:** Fatima Abdul Raheem (CEO), `fatima.abdulraheemdev.17@gmail.com`, is the committing GitHub account for the `nurvexthink` org.
- **Commits:** Conventional Commits (`feat:`, `fix:`, `chore:`, `docs:`).

7.1 Vercel/GitHub deployment notes (researched)

- A Vercel project name can differ from the GitHub username and repo name — no limitation.
- Vercel Hobby cannot deploy private repos owned by a GitHub org, and only the owner's commits trigger deploys → resolved by public repo + GitHub Actions (chosen), or by upgrading to Vercel Pro Team (~\$20/seat/mo) later.
- Cloudflare Pages is the fallback host (more generous org/team free tier, rougher Next.js support).

8 Org management

8.1 GitHub org (`nurvexthink`)

- Repos: `nurvexthink-web` + one repo per published product.
- Roles: CEO + shared `nurvexthink@gmail.com` = Owners; CFO = Owner or Maintainer.
- Require 2FA org-wide; enable Dependabot + secret scanning.
- Protect `main` with required PR review.

8.2 Supabase org (`nurvexthink`)

- One Supabase project for the main site; each product needing a DB gets its own project.
- Invite all 3 admins; Owners for CEO + shared account, Developer/Admin for CFO.
- Split into staging + prod projects when it matters (start with one).
- Daily backups (paid tier) when justified.

9 Performance & quality (researched best practices)

9.1 Next.js

- Server Components by default; keep `"use client"` at the leaves.
- `next/image` with `priority` for the above-the-fold hero (LCP).
- Lean on the React Compiler (Next 15+); cache/revalidate `fetch`.
- Target Core Web Vitals (LCP, INP, CLS).

9.2 React Three Fiber / Three.js

- `frameLoop="demand"` — render only on change.
- Mutate via `useFrame` + `refs`, never React state in the loop.
- Compressed textures (KTX2/Basis), compressed models (GLB + Draco), `useGLTF.preload()`.
- LOD via `drei <Detailed>`; dispose GPU resources; lazy-load 3D so it never blocks first paint.
- Monitor with `r3f-perf`.

9.3 Supabase / Postgres

- Index columns used in RLS policies and frequent filters.
- Test RLS with Studio impersonation before launch.

10 Testing

- Unit/component tests for form validation and admin auth logic.
- RLS policy tests (Studio impersonation, or `supabase test db` with `pgTAP`).
- Lighthouse / Core Web Vitals check before launch.
- Manual: lead form submits → row in `orders`; admin publishes a product/post → appears publicly.

11 Build order (high level)

1. Repo + tooling: Next.js + TS + Tailwind + shadcn, CLAUDE.md, skills, CI.
2. Brand/theme system + layout shell (nav, footer, dark theme).
3. Supabase project + schema + RLS + seed data.
4. Public pages: Home (with 3D hero), Products, Blog, Order form.
5. Admin auth + dashboard CRUD (products, blog, orders).
6. 3D hero polish + performance pass.
7. Deploy pipeline (GitHub Actions → Vercel) + domain.

12 Open items / future phases

- Client accounts + order tracking.
- Stripe payments.
- Staging Supabase environment.
- Per-product repos and deployments (separate specs).